

An AI-Based Intelligent Farming System for Improving Pomegranate Yield and Quality

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Pomegranate Fruit Growth Stage Detection System

Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT (Maximum 300 words)

Pomegranate cultivation is important due to its high nutritional value and economic benefits. However, farmers face challenges in monitoring fruit growth stages and identifying the correct harvesting time. Traditional methods rely on visual observation and experience, which are often inaccurate and may lead to reduced fruit quality and financial losses. Existing studies have applied computer vision and machine learning techniques for tasks such as disease detection, fruit grading, and object detection. However, these approaches mainly focus on post-harvest analysis or classification and do not support continuous monitoring of fruit growth stages, harvest time prediction, or stage-based decision-making for farmers. This creates a research gap in providing a complete, farmer-oriented solution.

To address this gap, this research proposes an AI-based system for pomegranate fruit growth stage detection and harvest prediction. The system allows farmers to upload fruit images through a web-based interface. It uses YOLO for fruit detection and Convolutional Neural Networks (CNN) for classifying the fruit into different growth stages. Based on the detected stage, a rule-based method is used to estimate the remaining time required for harvesting. Additionally, the system provides stage-based care tips and risk warnings to support better crop management.

The expected outcome of this system is improved accuracy in growth stage detection, better harvesting decisions, and increased crop productivity. It also helps reduce losses and improve fruit quality. Socially, the system supports farmers by providing easy-to-use technological assistance. Commercially, it can be developed as a low-cost smart farming platform for wider agricultural use.

Keywords: Pomegranate, Growth Stage Detection, Harvest Prediction, Computer Vision, Smart Farming

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
YOLO	You Only Look Once
ML	Machine Learning
IoT	Internet of Things
UI	User Interface
SDG	Sustainable Development Goal

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1. INTRODUCTION

1.1 Background and Context

Pomegranate is a highly nutritious fruit that is widely cultivated due to its economic value and health benefits. The global demand for pomegranates has increased significantly because they are rich in antioxidants and widely used in food, medicine, and beverage industries [1]. However, successful pomegranate cultivation requires careful monitoring of fruit growth stages, plant health, and environmental conditions to ensure high yield and quality. In traditional farming practices, farmers mainly rely on visual observation and personal experience to determine fruit maturity and harvesting time. This approach is often subjective and may lead to incorrect decisions, reducing fruit quality and market value [2].

In Sri Lanka, pomegranate cultivation is still developing, and a large portion of the demand is fulfilled through imports from other countries [3]. Therefore, improving local cultivation practices and supporting farmers with modern technological solutions is essential to increase production. One of the major challenges faced by farmers is identifying and monitoring the different growth stages of pomegranate fruits, from bud formation to full maturity. This growth period typically takes about five to seven months and requires continuous observation to ensure proper development and timely harvesting [4].

Recent studies have explored the use of computer vision and machine learning techniques in agriculture. For instance, Mitkal and Jagadale (2023) developed a system using image processing and Convolutional Neural Networks (CNN) to detect diseases and grade pomegranate fruits based on visual features such as color, texture, and shape [5]. While this approach provides useful insights for disease detection and quality assessment, it mainly focuses on fruit condition rather than monitoring the complete growth process or predicting harvest time.

To address these limitations, this research proposes an AI-based system for monitoring pomegranate fruit growth stages using images captured by farmers. The system analyzes fruit images to identify the current growth stage and estimates the remaining time required for harvesting. In addition, it provides basic recommendations to support better fruit development. This approach enables farmers to monitor fruit growth more effectively and make informed decisions throughout the cultivation period, ultimately improving productivity and fruit quality.

1.2 Literature Review

Pomegranate cultivation plays an important role in fruit agriculture due to its nutritional value and economic benefits. In Sri Lanka, pomegranate farming is mainly practiced using traditional methods, where farmers manually monitor fruit development and determine harvesting time based on visual observation and personal experience. Farmers typically examine characteristics such as size, color, and external appearance to estimate maturity. However, this manual process is time-consuming and highly subjective, and incorrect harvesting decisions can reduce fruit quality and market value [2]. Since pomegranate fruits require approximately five to seven months to mature from the bud stage, continuous monitoring of growth stages is essential for determining the correct harvesting time and maintaining fruit quality [6].

Recent research has explored the use of computer vision and deep learning techniques in agriculture. Mitkal and Jagadale [5] proposed a convolutional neural network (CNN)-based system for pomegranate disease detection and fruit grading, using image preprocessing, segmentation, feature extraction, and classification based on color, shape, and texture. Although the model achieved good results, it focuses mainly on fruit health classification and does not consider full growth stage monitoring or harvest prediction. Similarly, Kumar et al. [8] applied image processing techniques to extract features and used machine learning algorithms such as K-Nearest Neighbor and Naïve Bayes for classification. While effective for quality analysis, their approach relies on manually extracted features and mainly targets post-harvest grading rather than monitoring fruit development in the field.

Advancements in object detection have also been applied in agriculture. Zhang et al. [9] introduced a YOLO-based framework for detecting pomegranate fruits in orchard environments, achieving high accuracy even under complex conditions. However, this method focuses only on fruit detection and does not provide information about growth stages or maturity. Reddy et al. [10] developed a transfer learning-based model to classify pomegranate growth stages using images of five stages: bud, flower, early fruit, mid-growth, and mature fruit. Although the model achieved high accuracy, it does not extend to harvest time prediction or provide stage-based recommendations. Other studies, including IoT-based approaches [11] and post-harvest management techniques [6], mainly focus on irrigation, storage, and productivity improvements rather than monitoring fruit development during cultivation. Similarly, labor-based studies [10] analyze harvesting efficiency but do not use machine learning techniques for fruit maturity analysis.

Overall, existing studies demonstrate the potential of artificial intelligence, machine learning, and computer vision in agriculture. However, most systems focus on specific tasks such as disease detection, fruit grading, or object detection, and do not provide a complete solution for monitoring fruit growth stages and predicting harvest time. Therefore, this study proposes a web-based system where farmers can upload images of pomegranate fruits to detect growth stages, estimate harvest time using rule-based logic, and receive stage-specific fertilizer recommendations. This approach aims to provide a practical and accessible solution to support better decision-making and improve crop quality

1.3 Research Gap

Although recent studies have shown that artificial intelligence and machine learning can support pomegranate image analysis, there are still several important gaps that remain unaddressed. The study on pomegranate growth stage detection using YOLOv8 and hybrid machine learning mainly focuses on identifying the fruit's current stage among five classes: Bud, Flower, Early-Fruit, Mid-Growth, and Ripe. Its main contribution is improving classification performance by combining YOLOv8 with stacking, and voting methods, where the best F1-score reached 95.0% [8]. However, this study stops at growth stage classification and does not extend the output into a practical farmer-support decision such as predicting the remaining days to harvest, warning about harvest delay, or providing stage-based care recommendations. In real agricultural practice, simply knowing the current stage is not enough for farmers to decide when to harvest or what action should be taken next [8].

Similarly, existing pomegranate quality analysis studies mainly concentrate on post-harvest grading. For example, one paper uses feature extraction with KNN and Naive Bayes to classify pomegranate quality grades, achieving its best result with at 96.347% accuracy [9]. While this is useful for sorting fruits after harvest, it does not help farmers during the growing period. It does not monitor fruit development over time, detect which growth stage the fruit is currently in, or estimate harvest readiness. Therefore, such approaches are limited because they support quality assessment after harvesting, rather than timely field-level decision making before harvest [9].

Another related study discusses manual harvesting productivity and labor estimation in pomegranate cultivation. It shows that pomegranates should be harvested within a limited period, and delays can affect productivity and management decisions [14]. However, this work is based on labor and time study techniques, not image-based fruit monitoring. It does not provide an automated way to visually detect fruit growth stages in the field or predict harvest timing from uploaded fruit images. This shows a practical gap between traditional harvesting knowledge and intelligent computer vision systems [6].

Therefore, what is clearly missing in current approaches is an integrated farmer-oriented system that goes beyond classification alone. Existing systems either focus on growth stage detection only [9], quality grading only [8], or manual harvesting management only [6]. None of these studies combine visual fruit-stage identification with harvest day prediction, risk awareness, and actionable care tips in one workflow. This makes current approaches insufficient for real-world farmer use, because farmers need not only recognition of the stage, but also guidance about what to do next and when to harvest.

The novelty of your proposed component is that it can bridge this gap by transforming stage detection into a more practical agricultural support tool. In your system, when a farmer uploads a pomegranate fruit image from any one of the five growth stages, the system can first detect the current stage, then estimate the remaining time to harvest, and finally provide preventive care tips or warning messages relevant to that stage. For example, if the fruit is in a middle growth stage, the system could warn the farmer about disease risk, delayed maturity, or the importance of monitoring fruit condition before the final harvest window. This is more creative and useful than a normal image classification model, because it connects computer vision output with field-level decision support.

This research gap is also supported by evidence from the papers uploaded. The YOLOv8 hybrid study explicitly evaluates classification performance using metrics such as precision, recall, accuracy, but its conclusion is centered on better detection and classification accuracy rather than harvest prediction or advisory support [8]. The quality analysis paper is limited to grading based on external appearance and is meant for post-harvest management [6]. The harvesting productivity paper emphasizes the importance of timely harvesting and proper planning, which indirectly supports the need for systems that can estimate harvest timing more intelligently. Taken together, these studies confirm that there is room for a new system that not only identifies the fruit stage but also helps the farmer make timely and profitable harvest decisions.

2. OBJECTIVE

2.1 Main Objective

To develop an AI-based system that can accurately detect the growth stage of pomegranate fruits from images, predict the remaining days for harvesting, and provide stage-specific care recommendations to support farmers in decision making and yield improvement.

2.2 Specific Objectives

- To classify pomegranate fruits into five growth stages (Bud, Flower, Early Fruit, Mid Growth, and Mature) using image processing techniques
- To detect and isolate the fruit from the background using object detection techniques
- To estimate the remaining number of days required for harvesting based on the identified growth stage
- To provide stage-based care tips and risk warnings (such as disease risk, irrigation needs, and nutrient requirements) to assist farmers
- To develop a user-friendly system where farmers can upload fruit images and receive instant results
- To evaluate the system performance using metrics such as accuracy, precision, recall, and F1-score under real-world conditions

3. METHODOLOGY

3.1 Research Approach

This study uses a Design Science research approach together with an experimental evaluation method. The Design Science approach is selected because this research focuses on developing a new AI-based system for pomegranate fruit growth monitoring and harvest prediction. The main goal of the system is to analyze fruit images and identify the current growth stage of the fruit, and then predict the remaining time required for harvesting. This system is designed to support farmers in making better decisions about when to harvest fruits and how to manage crop growth effectively.

The experimental method is applied to test the performance of the proposed system. A dataset of pomegranate fruit images representing different growth stages is collected and used for training and testing the AI models. The trained models analyze the visual characteristics of fruit images and classify them into different growth stages.

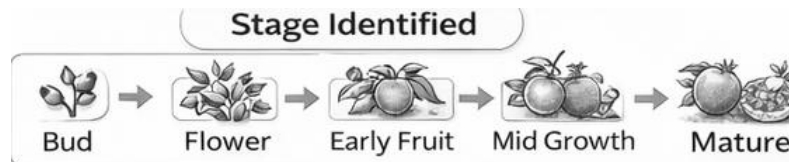


Figure 1: Pomegranate Fruit Growth Stages

To evaluate system performance, the results are measured using performance metrics such as classification accuracy and prediction reliability. This evaluation helps to determine how effectively the system can identify growth stages and support farmers in improving harvesting decisions.

3.2 Technical Methods and Algorithms

The proposed system applies image processing, deep learning, to analyze pomegranate fruit images and predict harvest time. First pomegranate fruit images are collected and organized into a dataset containing five growth stages: Bud, Flower, Early Fruit, Mid Growth, and Mature stage. Before training the models, the images are preprocessed using techniques such as image resizing, normalization, and background removal. These steps help improve image quality and ensure consistency in the dataset.

After preprocessing, deep learning models are used to analyze the images. A YOLO (You Only Look Once) model is used for fruit detection, which helps identify and isolate the fruit from the background. Then, a Convolutional Neural Network (CNN) model is used for feature extraction and classification. The CNN model learns important visual features such as color, size, shape, and texture that represent different growth stages.

Once the growth stage is identified, the system applies to a rule-based prediction method using predefined growth timelines to estimate the remaining days required for harvesting. Additionally, the system provides stage-based care tips and risk warnings, such as irrigation needs and disease prevention suggestions. The trained models are evaluated, and the model with the best performance is selected as the final model for the system.

3.3 Tools and Platforms with Justification

The proposed system uses several tools and technologies to support image processing, machine learning, and mobile application development. Python is used as the main backend programming language due to its strong support for artificial intelligence and availability of powerful libraries. For building and training the model, TensorFlow is used, as it supports deep learning techniques such as Convolutional Neural Networks (CNN), which are suitable for image classification tasks.

For object detection, the YOLO (You Only Look Once) framework is applied because it provides fast and accurate detection of fruits in images. This helps the system to identify and focus on the relevant fruit area. OpenCV is used for image preprocessing tasks such as resizing, normalization, and noise reduction, which improve image quality before model training. The system is developed as a mobile application using Flutter, allowing cross-platform usage on Android. Farmers can easily upload images and receive results through the app. The Flutter frontend is connected to the Python backend using APIs for smooth communication. These tools are selected because they are open-source, efficient, and suitable for developing AI-based mobile applications.

3.4 Data Collection and Ethical Considerations

The dataset used in this study consists of images of pomegranate fruits representing different growth stages. The images are collected from public datasets and agricultural sources. Additionally, some images may be collected from farms to include real-world conditions.

The dataset includes images of five growth stages: Bud, Flower, Early Fruit, Mid Growth, and Mature stage. Each image is labeled according to its corresponding growth stage before being used for training and testing. Ethical considerations are followed during data collection. Only fruit images are used, and no personal information about farmers is collected or stored. If images are collected from farms, proper permission is obtained before using them for research purposes. All data is used strictly for academic research and system development.

3.5 Validation Strategy and Performance Metrics

The performance of the proposed system is evaluated using a training and testing approach. The dataset is divided into two parts, where 80% of the images are used for training and 20% are used for testing. To evaluate the effectiveness of the system, several performance metrics are used. Accuracy is used to measure how correctly the system classifies fruit images into their respective growth stages. The target is to achieve high classification accuracy. In this research, deep learning models such as YOLO and CNN are used and evaluated. The system performance is tested using new unseen fruit images to ensure reliability. In addition to classification accuracy, the effectiveness of the harvest time prediction is also evaluated based on how closely the predicted days match the expected growth timelines. The model that provides the best performance in both growth stage detection and harvest prediction is selected as the final model for the system.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed solution is an AI-based intelligent farming system developed to improve pomegranate yield, harvesting efficiency, and crop management. Many farmers currently depend on manual observation and experience to decide the correct growth stage and harvesting time, which can lead to early or late harvesting and reduced fruit quality. To overcome this problem, the system is designed as a mobile platform that supports farmers throughout different stages of pomegranate cultivation.

The overall system includes four main components. The first component is Soil Health Analysis and Smart Fertilizer & Water Recommendation, which monitors soil conditions and provides fertilizer and irrigation suggestions. The second component is Fruit Growth Monitoring and Harvest Prediction, which identifies fruit growth stages and estimates the remaining time for harvesting. The third component is Disease Detection and Treatment Recommendation, which detects fruit diseases and provides suitable treatment guidance. The fourth component is Fruit Quality Grading and Waste Utilization Analysis, which analyzes fruit quality after harvesting.

The Fruit Growth Monitoring and Harvest Prediction component, which is the focus of this research, works during the pre-harvest stage. It analyzes fruit images uploaded by farmers, detects the current growth stage, and predicts the approximate number of days remaining until harvest. In addition, the system provides stage-based care guidance and risk warnings, helping farmers take timely actions to improve fruit development and prevent potential issues.

By combining all components, the system supports farmers from soil preparation to harvesting and post-harvest management. The interaction between these components enables continuous monitoring and decision support, helping farmers increase productivity, reduce losses, and improve overall pomegranate farming outcomes.

4.2 System diagram of components

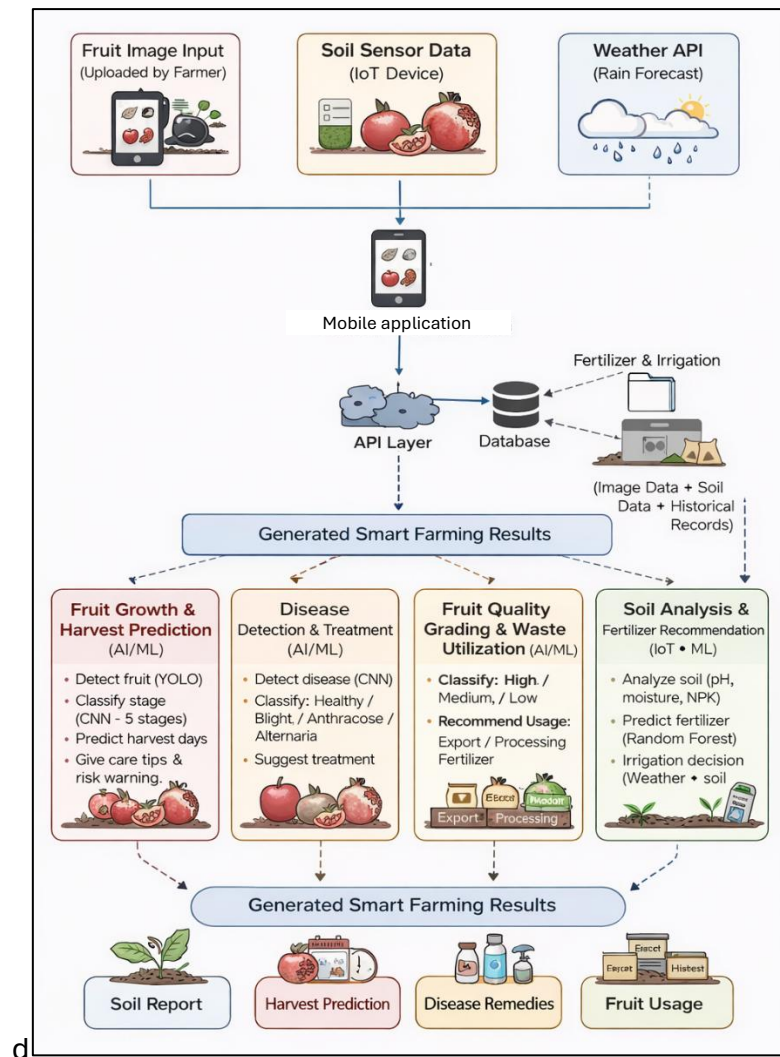


Figure 2: Data flow of System Diagram of component

4.3 Highlight of the Individual Component and Its Parts

This research mainly focuses on the Fruit Growth Monitoring and Harvest Prediction component. This component is designed with several functional modules that work together to identify the fruit growth stage, estimate harvest time, and provide care guidance. First, the image upload module allows farmers to upload pomegranate fruit images using the web system. Next, the image preprocessing module prepares the image by resizing, normalizing, and removing background noise to improve image quality. After preprocessing, the image is passed to the fruit detection module, where a deep learning model such as YOLO detects and isolates the fruit from the image.

Then, the processed image is sent to the growth stage classification module, where a Convolutional Neural Network (CNN) analyzes the visual features of the fruit, such as color, size, shape, and texture. Based on these features, the system classifies the fruit into five growth stages: Bud, Flower, Early Fruit, Mid Growth, and Mature stage.

After classification, the result is passed to the harvest prediction module, which estimates the remaining number of days required for harvesting using predefined growth timelines and rule-based logic. In addition, the care guidance module provides stage-based recommendations such as irrigation needs, fertilizer suggestions, and risk warnings to support proper crop management. Finally, the result display module presents the output to the farmer through the web interface, including the detected growth stage, estimated days to harvest, and recommended care actions.

4.4 Component Interactions

Although this component focuses on fruit growth stage detection and harvest prediction, it relates to other parts of the system. The Soil Health Analysis component provides information about soil conditions such as nutrients and moisture levels, which can influence fruit growth and development. This information can help farmers take appropriate actions to improve crop growth. The Disease Detection component also supports this module by identifying any diseases affecting fruit. If a fruit is infected, it may impact the growth stage and delay the harvesting time. This information can be used to adjust the harvest prediction and provide additional care recommendations. In this way, the outputs from other components enhance the accuracy of growth stage detection and harvest prediction, providing better decision support to farmers.

4.5 Data Flow Explanation

The data flow of this component starts when the user uploads a pomegranate fruit image through the web interface. First, the image is sent to the preprocessing stage, where it is resized, normalized, and cleaned to improve image quality. Next, the processed image is passed to the fruit detection stage, where the system identifies and extracts the fruit region from the image.

After that, the image is sent to the growth stage classification stage. Here, the system extracts visual features such as color, size, and texture and applies deep learning models to classify the fruit into one of the five growth stages: Bud, Flower, Early Fruit, Mid Growth, and Mature stage. Once the growth stage is identified, the result is passed to the harvest prediction stage. The system uses predefined growth timelines and rule-based logic to estimate the remaining number of days required for harvesting. In addition, weather-related factors such as rainfall can be used to adjust the predicted harvest time. Finally, the system generates care recommendations based on the detected stage and displays the output to the user. The output includes the current growth stage, estimated days to harvest, and suggested care actions, which are presented through the web interface.

4.6 Scalability and Security Considerations

The proposed component is developed as part of a web-based system, allowing farmers to access it from different locations using a browser. As the number of users increases, the system can be easily expanded by using cloud-based services and larger storage capacity. This enables the system to handle more uploaded fruit images and perform growth stage analysis and harvest prediction efficiently in the future. Scalability can also be improved by enhancing the dataset with more fruit images from different environments. This helps the model learn better and improves the accuracy of growth stage detection and harvest prediction over time. Security is also an important consideration in this component. User authentication and login mechanisms can be implemented to ensure that only authorized users can access the system. Uploaded fruit images and prediction results should be stored securely in the database. In addition, secure communication protocols should be used to protect data during transmission between the user and the web application. These considerations ensure that the component is reliable, scalable, and secure, making it suitable for real-world agricultural applications.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed system involves several stakeholders who interact with the system or benefit from its results. The main users of the system are pomegranate farmers. They can upload images of pomegranate fruits during different growth stages using the web system. The system will analyze the fruit image, identify the current growth stage, and predict the remaining time required for harvesting. It also provides tips and warnings, helping farmers make better decisions during cultivation.

Another stakeholder group is farm owners and agricultural managers. They can use the system to monitor fruit development and plan harvesting schedules more effectively. This helps improve productivity and reduce losses due to incorrect harvesting time. Agricultural officers and researchers can use the system data to study fruit growth patterns, harvesting timelines, and environmental impacts. This supports better agricultural planning and research.

System administrators are responsible for managing and maintaining the system. They manage the dataset, update AI models, and ensure the system operates correctly. Therefore, the main stakeholders of this system are:

- Pomegranate farmers (primary users)
- Farm owners and agricultural managers
- Agricultural officers and researchers
- System administrators

5.2 Functional Requirements

The proposed system shall provide the following functions:

FR1: The system shall allow users to upload pomegranate fruit images through a web-based interface.

FR2: The system shall preprocess the uploaded image to improve image quality.

FR3: The system shall detect the fruit from the image using an AI-based detection model.

FR4: The system shall classify the fruit into one of five growth stages: Bud, Flower, Early Fruit, Mid Growth, or Mature.

FR5: The system shall predict the estimated number of days remaining until harvest.

FR6: The system shall provide stage-based care recommendations and risk warnings.

FR7: The system shall display the detected growth stage, harvest time, and care tips to the user.

FR8: The system shall allow users to view the analysis results through the web interface.

5.3 Non-Functional Requirements

Non-functional requirements describe how the system should perform in real usage conditions.

Performance (Speed)

The system should analyze a pomegranate fruit image and provide results within 2–3 seconds after the image is uploaded, enabling quick decision-making for farmers.

Accuracy

The AI model should achieve at least 90% accuracy in correctly identifying the fruit growth stage and providing reliable harvest predictions.

Usability

The system should have a simple and user-friendly web interface so that farmers can easily upload images and understand the results without technical knowledge.

Reliability

The system should provide consistent and stable results when similar fruit images are analyzed, ensuring trust in the system's predictions.

Maintainability

The system should allow easy updates to the dataset and AI models in the future to improve performance and adapt to new data.

6. COMMERCIALIZATION PLAN

The proposed AI-Based Intelligent Farming System for improving pomegranate yield and quality can be commercialized as a low-cost, web-based smart farming platform in Sri Lanka. The main target market includes pomegranate farmers, agricultural farms, cooperatives, exporters, and agricultural officers. The primary customers are small and medium-scale farmers who need an easy and affordable solution to manage soil conditions, detect diseases early, monitor fruit growth, and improve grading accuracy. The system provides strong value by combining four main components: soil monitoring with fertilizer and irrigation recommendations, fruit growth stage detection with harvest time prediction, fruit disease detection with treatment suggestions, and fruit quality grading with waste utilization recommendations. This gives a complete solution from pre-harvest to post-harvest decision making.

The system can generate revenue through a monthly subscription model of around LKR 2,000–3,000, selling IoT soil monitoring devices (approximately LKR 8,000 per unit), and through partnerships with government and agricultural organizations. The overall cost includes AI model development, web application development, IoT device setup, cloud hosting, and system maintenance, but these costs can be managed and scaled over time. A flexible pricing strategy such as a freemium model and bundled packages helps make the system affordable for farmers.

This system has a strong competitive advantage because it integrates all four agricultural components into one platform, focuses on fruit-based analysis instead of only leaf-based detection, and provides practical recommendations rather than just showing data. In addition, intellectual property can be protected through copyrights, patents, and licensing of the AI models, IoT integration, and the overall system design. Overall, this commercialization plan shows that the system is practical, affordable, and useful for improving farmer productivity and income.

7. BUDGET AND JUSTIFICATION

Category	Item	Estimated Cost (LKR)	Justification
Hardware Costs	Laptop Maintenance / Service	6,000	Required to ensure smooth system development, model training, and testing.
Travel Costs	Field Visit (Jaffna & Horana Farms)	20,000	Required to collect real-world pomegranate fruit images and gain practical farming knowledge.
Cloud Services	Cloud Hosting & Storage	5,000	Used to deploy the web system and store uploaded images and results.
Internet Costs	Data Usage for Training & Upload	5,000	Required for dataset downloading, model training, and system testing.
Human Effort	Development & Research Effort	10,000	Includes time spent on coding, dataset preparation, training models, and system development.
TOTAL		46,000	Overall estimated cost for system development and deployment.

Table 1: Estimated Budget for Disease Detection Component

8. WORK BREAKDOWN STRUCTURE (WBS)

Task breakdown per member

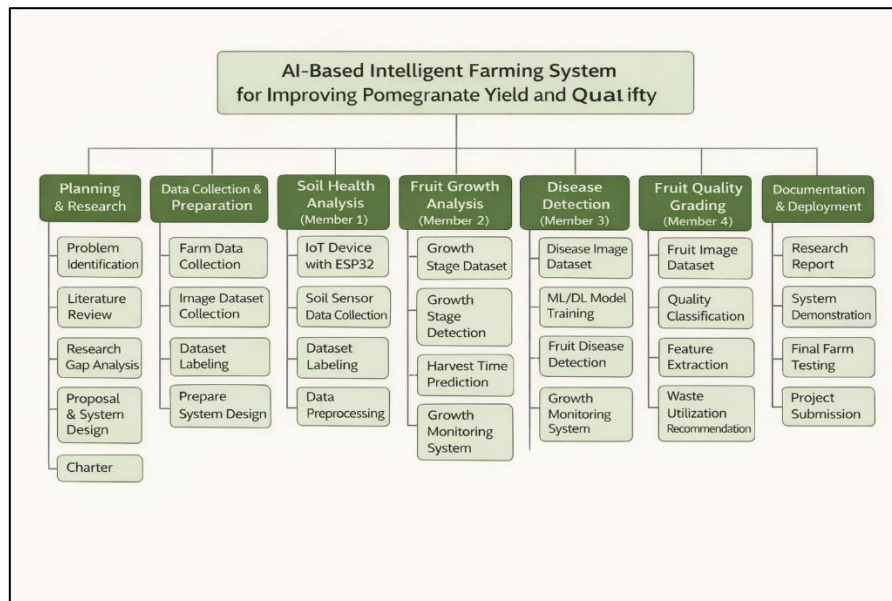


Figure 3: Task Breakdown

Timeline alignment



Figure 4: Timeline Alignment

Balanced workload distribution

The workload was evenly distributed among all team members by assigning each member a specific component of the system. Each component had similar complexity and required equal effort in terms of data collection, model development, and implementation. In addition to individual tasks, all members contributed to shared activities such as system integration, testing, and report writing. Regular discussions and progress meetings were conducted to ensure that the work was balanced and completed on time.

9. GANTT CHART



Figure 5: Gantt Chart

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11.APPENDICES

Appendix A: Farmer Interviews

Interviews were conducted with a small group of farmers to understand the challenges in pomegranate cultivation, especially related to fruit growth monitoring and harvesting decisions. The aim was to identify how farmers currently track fruit development stages and determine the correct time for harvesting. The following questions were used to collect information from farmers

Interview Questions:

- What growth stages of pomegranate fruits do you usually observe on your farm (bud, flower, early fruit, mid growth, mature)?
- How do you currently identify the growth stage of the fruit?
- Do you face difficulties in understanding whether the fruit is developing normally or not?
- How do you currently decide the correct time to harvest pomegranate fruits?
- Have you ever experienced losses due to early or late harvesting? If yes, how often?
- Do you monitor fruit growth regularly during the cultivation period? If yes, how?



Figure 6:Collecting Pomegranate Fruit in Field



Figure 7:Interview with farmer for Data Collection

Appendix B: Detailed diagrams

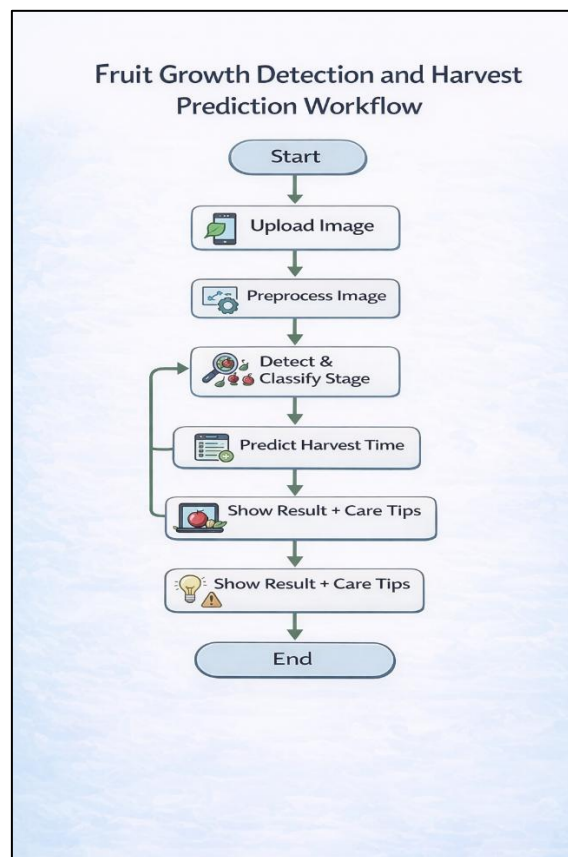


Figure 8: Fruit Growth Detection and Harvest predication workflow diagram

Appendix C: Preliminary UI mockups

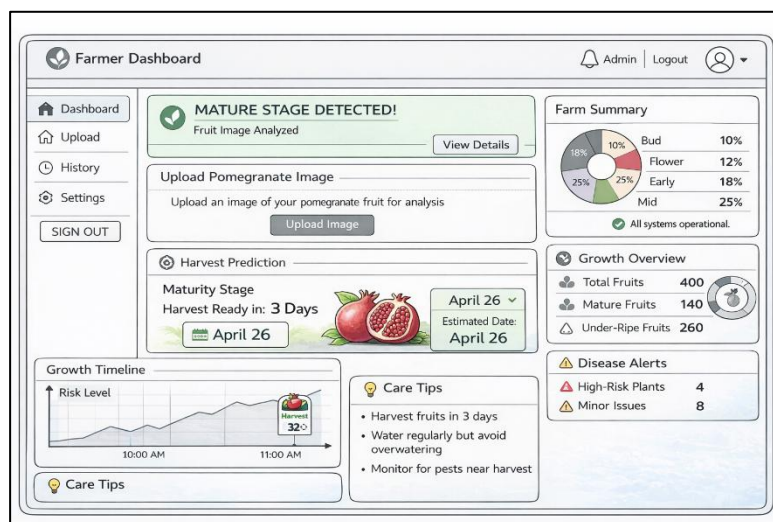


Figure 9:: Fruit growth detection and harvest predication dashboard

Appendix D: Customer Personas

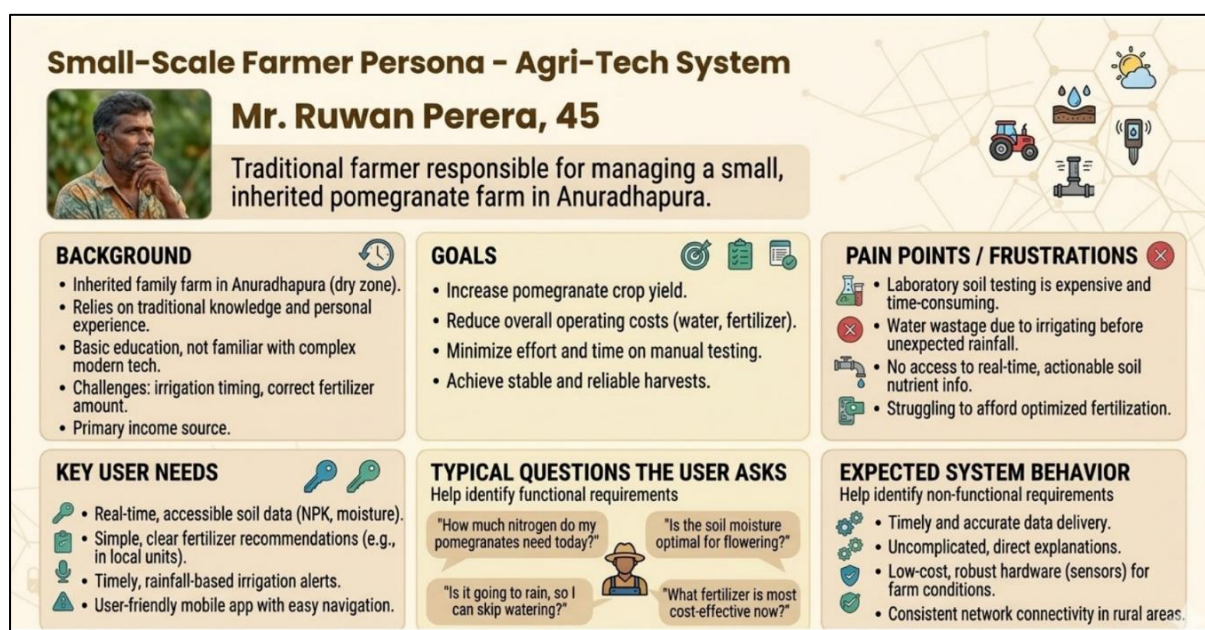


Figure 10: Small Scale farmer persona